

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Ethanol 95% v/v

**CAS No** 64-17-5

**Synonyms** Ethyl alcohol

**Product Code** E/0500DF/08; E/0500DF/15; E/0500DF/17; E/0500DF/21; E/0500/DF25; E/0500DF/C17; E/0500DF/P21; E/0500DF/P24; E/0500DF/P25; E/0500DF/21S

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Serious Eye Damage/Eye Irritation

Category 2

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Exclamation Mark

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor

H319 - Causes serious eye irritation

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P241 - Use explosion-proof electrical/ ventilating/ lighting equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P260 - Do not breathe dust/fume/gas/mist/vapors/spray

P264 - Wash face, hands and any exposed skin thoroughly after handling

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear eye protection/ face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P308 + P313 - IF exposed or concerned: Get medical advice/attention

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

This product does not contain any known or suspected endocrine disruptors

## Section 3 - Composition and Information on Ingredients

| Component     | CAS No  | Weight % |
|---------------|---------|----------|
| Ethyl alcohol | 64-17-5 | 95-96    |

## Section 4 - First Aid Measures

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

|                                            |                                                                                                                                              |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                       |
| <b>Self-Protection of the First Aider</b>  | Remove all sources of ignition.                                                                                                              |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                         |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                              |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Specific Hazards Arising from the Chemical

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Vapors may form explosive mixtures with air.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

| Component     | Australia                                    | New Zealand WEL                                                                             | ACGIH TLV      | The United Kingdom                                                                                               | Germany                                           |
|---------------|----------------------------------------------|---------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Ethyl alcohol | TWA: 1000 ppm<br>TWA: 1880 mg/m <sup>3</sup> | TWA: 200 ppm<br>TWA: 380 mg/m <sup>3</sup><br>STEL: 800 ppm<br>STEL: 1520 mg/m <sup>3</sup> | STEL: 1000 ppm | TWA: 1000 ppm TWA;<br>1920 mg/m <sup>3</sup> TWA<br>WEL - STEL: 3000 ppm<br>STEL; 5760 mg/m <sup>3</sup><br>STEL | 200 ppm TWA MAK;<br>380 mg/m <sup>3</sup> TWA MAK |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness   | AUS/NZ Standard | Glove comments                                                                 |
|----------------|-------------------|-------------------|-----------------|--------------------------------------------------------------------------------|
| Butyl rubber   | > 480 minutes     | 0.38 mm - 0.56 mm | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene       | > 480 minutes     | 0.45 mm           |                 |                                                                                |
| Viton (R)      | > 480 minutes     | 0.30 mm           |                 |                                                                                |
| Nitrile rubber | < 60 minutes      | 0.18 mm           |                 |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger

of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                                                                       |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                                                                  |
| <b>Recommended half mask:-</b>         | Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.                                                                                                                                                                                                                       |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system.                                                                                                                                                                                                                             |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                                 |                                             |
|------------------------------------------------|-------------------------------------------------|---------------------------------------------|
| <b>Appearance</b>                              | Clear, Colorless                                |                                             |
| <b>Physical State</b>                          | Liquid                                          |                                             |
| <b>Odor</b>                                    | sweet, Characteristic                           |                                             |
| <b>Odor Threshold</b>                          | No data available                               |                                             |
| <b>pH</b>                                      | No information available                        |                                             |
| <b>Melting Point/Range</b>                     | -114 °C / -173.2 °F                             |                                             |
| <b>Softening Point</b>                         | No data available                               |                                             |
| <b>Boiling Point/Range</b>                     | 78 °C / 172.4 °F                                |                                             |
| <b>Flash Point</b>                             | 13 - 17 °C / 55.4 - 62.6 °F                     | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available                               |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                                  | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 3.3 vol %<br><b>Upper</b> 19 vol % |                                             |
| <b>Vapor Pressure</b>                          | No data available                               |                                             |
| <b>Vapor Density</b>                           | No data available                               | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.80                                            |                                             |
| <b>Bulk Density</b>                            | Not applicable                                  | Liquid                                      |
| <b>Water Solubility</b>                        | Soluble                                         |                                             |
| <b>Solubility in other solvents</b>            | No information available                        |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                 |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                  |                                             |
| Ethyl alcohol                                  | -0.32                                           |                                             |
| <b>Autoignition Temperature</b>                | 363 °C / 685.4 °F                               |                                             |
| <b>Decomposition Temperature</b>               | No data available                               |                                             |
| <b>Viscosity</b>                               | No data available                               |                                             |
| <b>Explosive Properties</b>                    |                                                 | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available                        |                                             |
| <b>Other information</b>                       |                                                 |                                             |
| <b>Molecular Formula</b>                       | C <sub>2</sub> H <sub>6</sub> O                 |                                             |
| <b>Molecular Weight</b>                        | 46.07                                           |                                             |

## Section 10 - Stability and Reactivity

|                                         |                                                                                          |
|-----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                                               |
| <b>Stability</b>                        | Stable under normal conditions.                                                          |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition, Incompatible products. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Strong acids, Acid anhydrides, Acid chlorides.                  |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                 |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                 |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

| Component     | LD50 Oral                                                    | LD50 Dermal | LC50 Inhalation                                                   |
|---------------|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| Ethyl alcohol | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) |             | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |

(b) skin corrosion/irritation; Based on available data, the classification criteria are not met

(c) serious eye damage/irritation; Category 2

##### (d) respiratory or skin sensitization;

**Respiratory**

Based on available data, the classification criteria are not met

**Skin**

Based on available data, the classification criteria are not met

| Component                          | Test method                                    | Test species | Study result    |
|------------------------------------|------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 95-96 ) | Mouse Ear Swelling Test (MEST)                 | mouse        | non-sensitising |
|                                    | OECD Test Guideline 429 Local Lymph Node Assay | mouse        | non-sensitising |

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

| Component                          | Test method                                   | Test species          | Study result |
|------------------------------------|-----------------------------------------------|-----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 95-96 ) | AMES test<br>OECD Test Guideline 471          | in vitro<br>Bacteria  | negative     |
|                                    | Gene cell mutation<br>OECD Test Guideline 476 | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; Based on available data, the classification criteria are not met

Ethanol has been shown to be carcinogenic in long-term studies only when consumed and abused as an alcoholic beverage.

(g) reproductive toxicity; Based on available data, the classification criteria are not met

| Component                          | Test method             | Test species / Duration   | Study result          |
|------------------------------------|-------------------------|---------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5 ( 95-96 ) | OECD Test Guideline 416 | Oral / mouse 2 Generation | NOAEL = 13.8 g/kg/day |
|                                    | OECD Test Guideline 414 | Inhalation / Rat          | NOAEC = 16000 ppm     |

|                                            |                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| (h) STOT-single exposure;                  | Based on available data, the classification criteria are not met                                                    |
| (i) STOT-repeated exposure;                | Based on available data, the classification criteria are not met                                                    |
| Target Organs                              | None known.                                                                                                         |
| (j) aspiration hazard;                     | Based on available data, the classification criteria are not met                                                    |
| Other Adverse Effects                      | The hazards associated with ethanol may be seen in this product.                                                    |
| Symptoms / effects, both acute and delayed | Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |

## Section 12 - Ecological Information

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component     | Freshwater Fish                                                  | Water Flea                                    | Freshwater Algae                              | Microtox                                                                                                                |
|---------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Photobacterium<br>phosphoreum: EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 =<br>35470 mg/L/5 min |

### Persistence and Degradability

#### Persistence

Persistence is unlikely, based on information available.

| Component                          | Degradability   |
|------------------------------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 95-96 ) | OECD 301E = 94% |

#### Bioaccumulative Potential

Bioaccumulation is unlikely

| Component     | log Pow | Bioconcentration factor (BCF) |
|---------------|---------|-------------------------------|
| Ethyl alcohol | -0.32   | No data available             |

#### Mobility

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility Disperses rapidly in air

#### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

#### Persistent Organic Pollutant

This product does not contain any known or suspected substance

#### Ozone Depletion Potential

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

#### Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

#### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### Other Information

Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations.

## Section 14 - Transport Information

**IMDG/IMO**

UN-No UN1170  
Proper Shipping Name ETHANOL  
Hazard Class 3  
Packing Group II

**ADG**

UN-No UN1170  
Proper Shipping Name ETHANOL  
Hazard Class 3  
Packing Group II

| Component                          | Hazchem Code |
|------------------------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 95-96 ) | 2YE<br>2Y    |

**IATA**

UN-No UN1170  
Proper Shipping Name ETHANOL  
Hazard Class 3  
Packing Group II

Environmental hazards No hazards identified  
Special Precautions No special precautions required  
Additional information None known

## Section 15 - Regulatory Information

**Safety, health and environmental regulations/legislation specific for the substance or mixture****National Regulations** Australia

See section 8 for national exposure control parameters.

**Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component               | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-------------------------|-------------------------------------------------------------|------------------------|
| Ethyl alcohol - 64-17-5 | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern



**National pollutant inventory** Subject to reporting requirements

| Component               | National pollutant inventory      |
|-------------------------|-----------------------------------|
| Ethyl alcohol - 64-17-5 | 10 tonne/yr. Threshold category 1 |

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

#### International Inventories

| Component     | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|---------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Ethyl alcohol | X    | X     | 200-578-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-13217 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Basel convention on the control of transboundary movements of hazardous wastes and their disposal

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component               | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|-------------------------|------------------------------------|------------------------------------------------------------------------|
| Ethyl alcohol - 64-17-5 | Annex I - Y42                      | Y42 except Halogenated solvents                                        |

| Component     | CAS No  | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|---------------|---------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethyl alcohol | 64-17-5 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH** Not applicable

## Section 16 - Other Information

#### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances

## Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances**PICCS** - Philippines Inventory of Chemicals and Chemical Substances**TWA** - Time Weighted Average**IARC** - International Agency for Research on Cancer**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association**MARPOL** - International Convention for the Prevention of Pollution from Ships**NZS 5433:2020** - Transport of Dangerous Goods on Land**LD50** - Lethal Dose 50%**EC50** - Effective Concentration 50%**WEL** - Workplace Exposure Limit**DNEL** - Derived No Effect Level**POW** - Partition coefficient Octanol:Water**vPvB** - very Persistent, very Bioaccumulative**VOC** - (Volatile Organic Compound)**KECL** - Korean Existing and Evaluated Chemical Substances**CAS** - Chemical Abstracts Service**ACGIH** - American Conference of Governmental Industrial Hygienists  
Predicted No Effect Concentration (PNEC)**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail**OECD** - Organisation for Economic Co-operation and Development**LC50** - Lethal Concentration 50%**ATE** - Acute Toxicity Estimate**RPE** - Respiratory Protective Equipment**NOEC** - No Observed Effect Concentration**BCF** - Bioconcentration factor**PBT** - Persistent, Bioaccumulative, Toxic

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

## Revision Date

07-Feb-2025

## Revision Summary

SDS sections updated, 2, 3, 4, 11, 12.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet